Amal Zulkipli

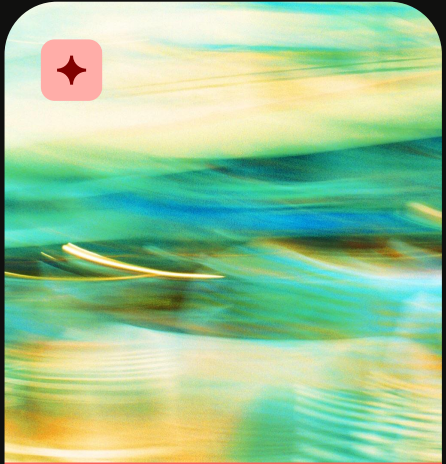
App Building & AI Literacy For Pharmacist

UniSZA Kampus Besut, Terengganu
27 October 2025 | 9:00 AM - 12:00 PM

What We'll Accomplish Today

Understand how apps actually work

Frontend · Backend · Database · Code

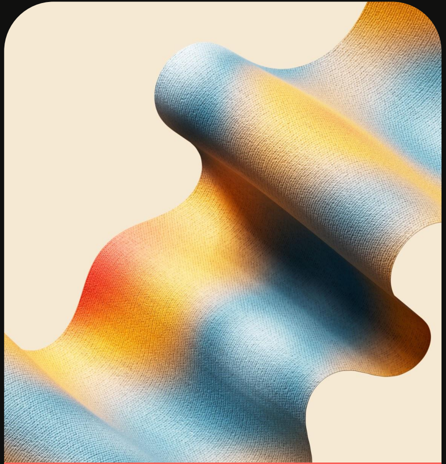


Learn to use AI tools for app building

LLM · ChatGPT · Prompt · AI Tools

Watch a live app being built from scratch

HTML · CSS · Build · Deploy



Build your own personal app prototype

Tailored · Curated

Topic 1

Understand how apps actually work

Quick Question?

Who has created apps/ website?
Written an automation scripts?
Learn to code?

Apps you use daily

MySejahtera	Digital tools for health tracking, appointment and announcement
WhatsApp	Digital tools for communication and promotion
Grab	Digital tools for transportation booking, food delivery, taxi and etc
Instagram	Digital tools for social media

Not all apps are created equal

Aspect	Mobile Apps	Web Apps	Desktop Apps
Definition	Applications installed on smartphones/tablets	Applications accessed through web browsers	Applications installed on computers (Windows/Mac/Linux)
Platform	iOS (Apple) or Android (Google)	Any device with a browser (Chrome, Safari, Firefox)	Windows, macOS, or Linux
Internet Requirement	Some work offline, most need internet	Usually requires internet connection	Can work fully offline
Updates	Must update through app store	Updates automatic (refresh browser)	Must download and install updates
Examples	MySejahtera, MIMS mobile, Medscape	MIMS online, PubMed, DrugBank	Microsoft Office, Excel, Word
Best For	On-the-go access, using phone features (camera, GPS)	Accessible from any device, easy to share	Complex tasks, processing large data, advanced features

How Does App work

Frontend
Counter

The face of your app

- What customers/user see
- Buttons, forms, display

Backend
Pharmacist

The brain of your app

- Processing & logic
- Calculations, decisions

Database
Storage

The memory of your app

- Where information lives
- Eg: Drug data, patient records

Understand how apps actually work

Today's Focus

Frontend

Counter

Backend

Pharmacist

Database

Storage

We will build **Frontend** + **Database** and skip **Backend** for now

Understand how apps actually work

8

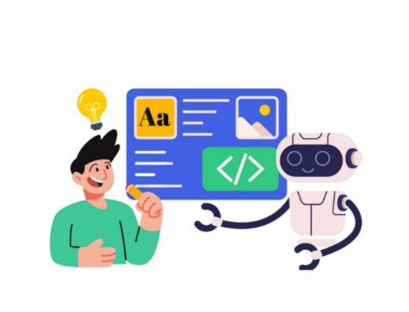
Old Way vs New Way

Traditional coding

Regular Coding

- Learn syntax
- Memorize commands
- Debug line by line

Plan → Code → Test → Iterate → Working App



AI-assisted coding

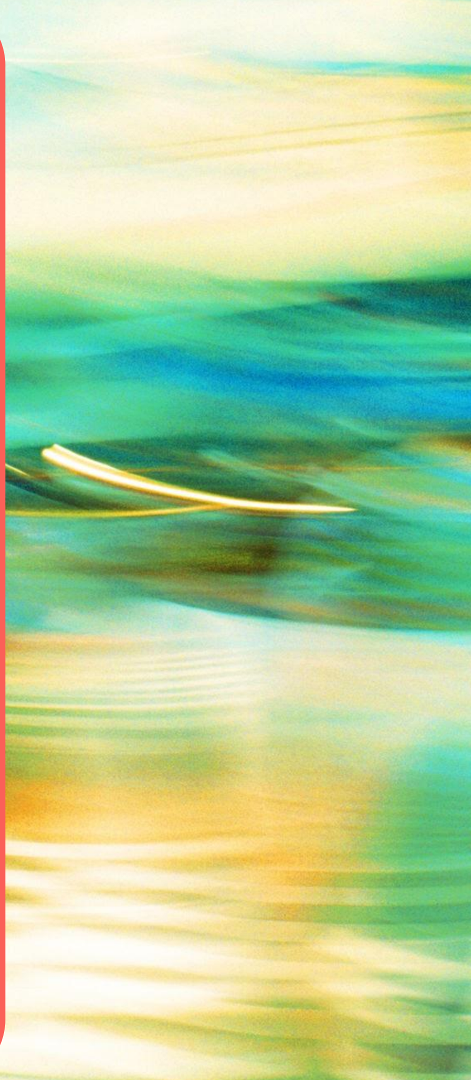
AI-Assisted Coding

- Describe your needs
- AI generates code
- Test and refine

Natural Language → Working App

Topic 2

Learn to use AI tools for app building



Quick Question?

Who has used ChatGPT, Claude, Gemini or any AI chat before?

What is actually AI?

Learn to use AI tools for app building

What is actually AI?

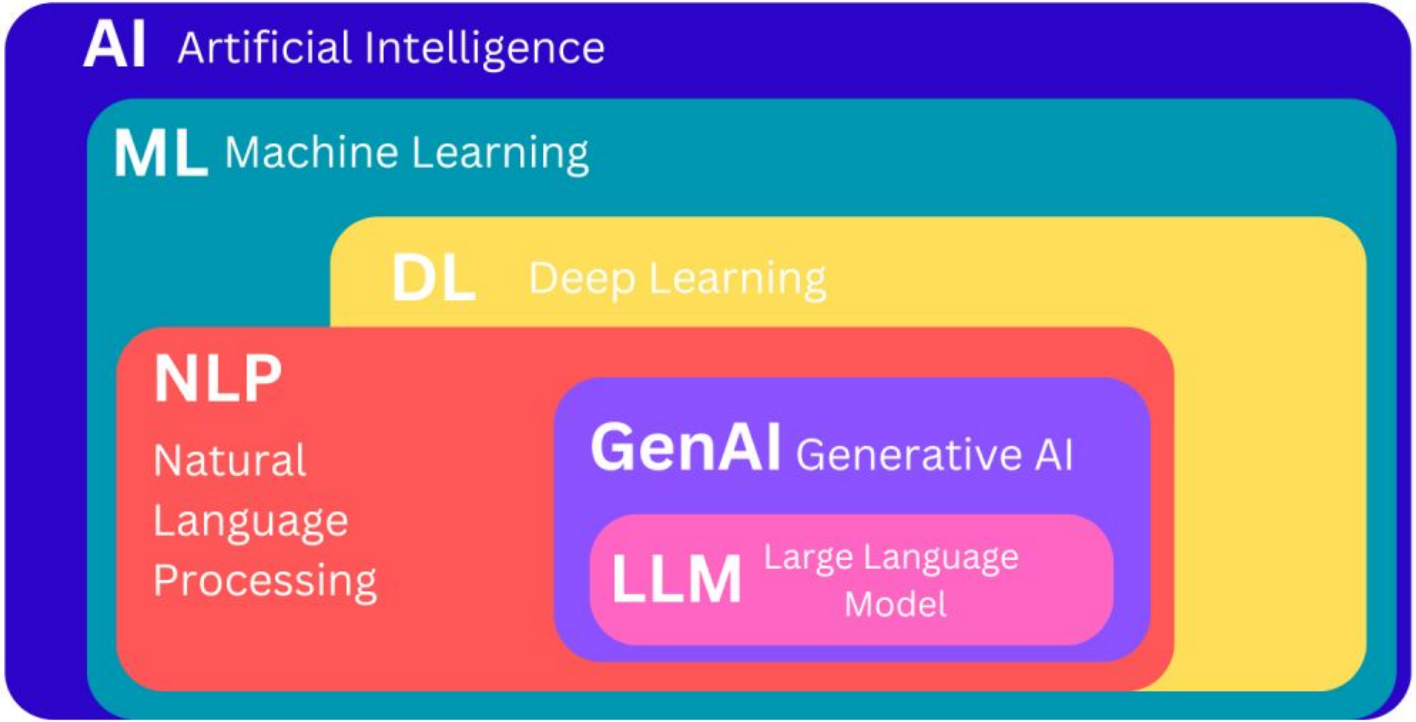
Artificial Intelligence

A mathematics and statistics applied at massive scale. It learns patterns from all existed data, just like you learn pharmacy by studying cases and textbooks.

Think of it like teaching computers to do tasks that normally require human intelligence

Examples you use daily:

1. Voice assistants (Siri, Google Assistant)
2. Face recognition (unlocking your phone)
3. Music recommendations (Spotify)
4. Navigation (Waze predicting traffic)
5. Drug Discovery (CADD)


AI Artificial Intelligence
ML Machine Learning
DL Deep Learning
NLP
Natural Language Processing
GenAI Generative AI
LLM Large Language Model

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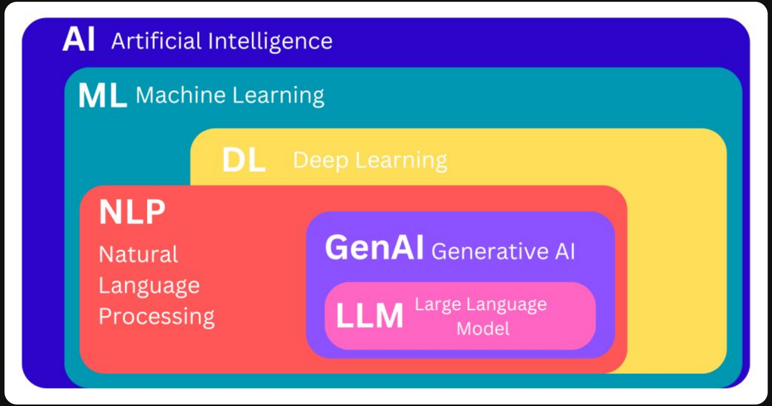
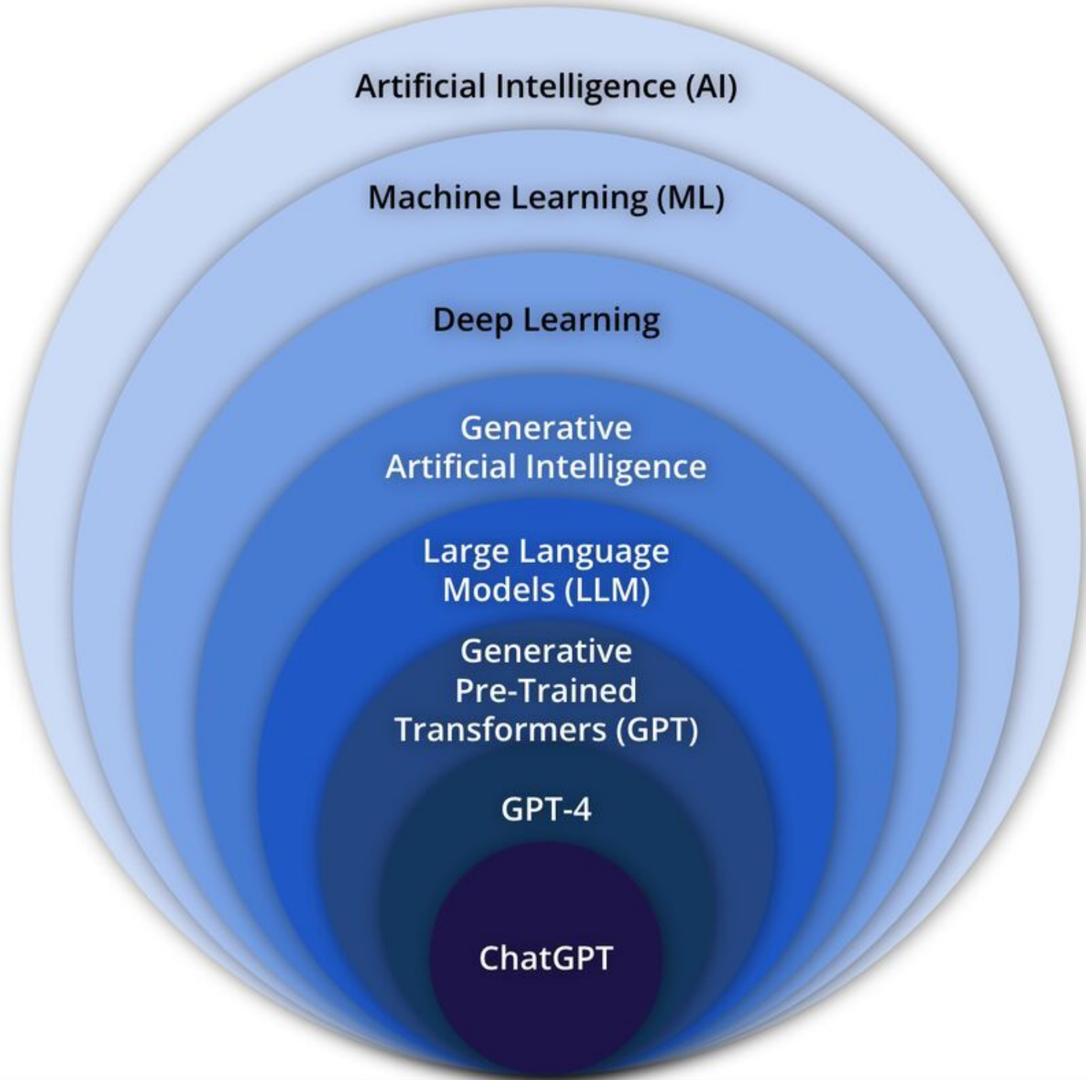
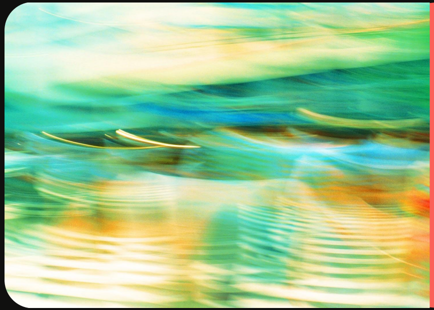




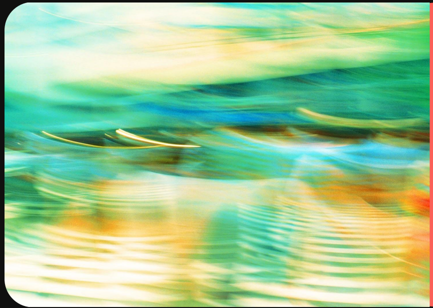
So what is LLM?

(Large Language Model)



Large Language Model

Trained on massive amounts of text to understand context, generate human-like responses and assist with complex tasks.



Large Language Model

Trained on massive amounts of text to understand context, generate human-like responses and assist with complex tasks.

Large

Trained on billions of text examples from the internet

Language

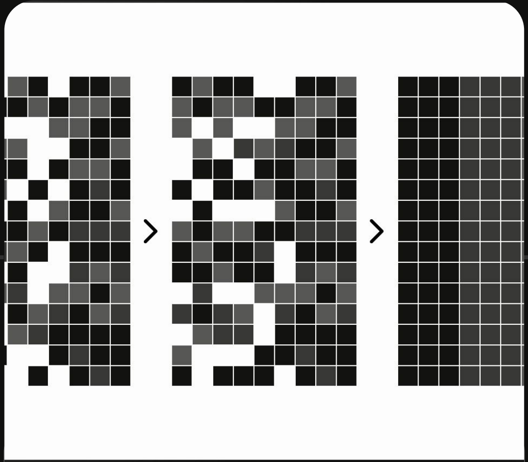
Understands and generates human language (English, Malay, code, etc.)

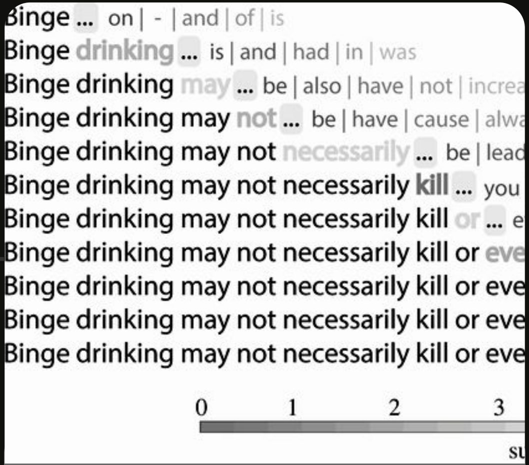
Model

A mathematical pattern that predicts what word/sentence comes next

Reading Phase

AI reads millions of books, websites, code repositories





Example:
"antibiotic" often appears near
"bacteria," "infection," "penicillin"

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Reading Phase

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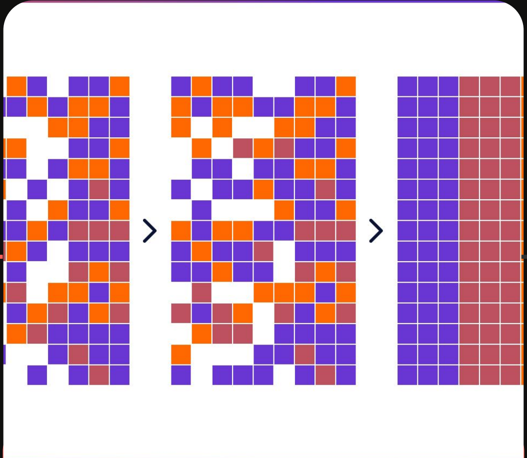
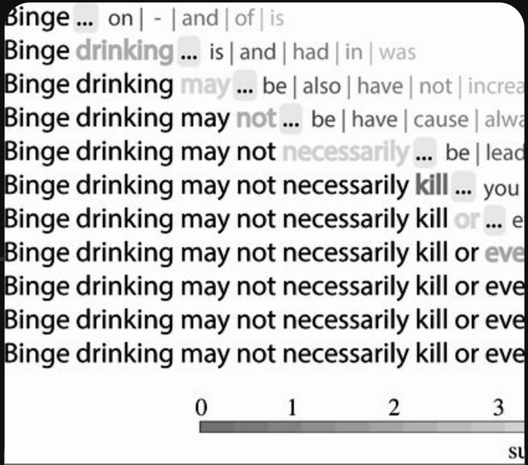
Pattern Recognition



Builds connections between concepts and understands context

Example:
Knows "500mg paracetamol" is a dosage, not a chemical formula



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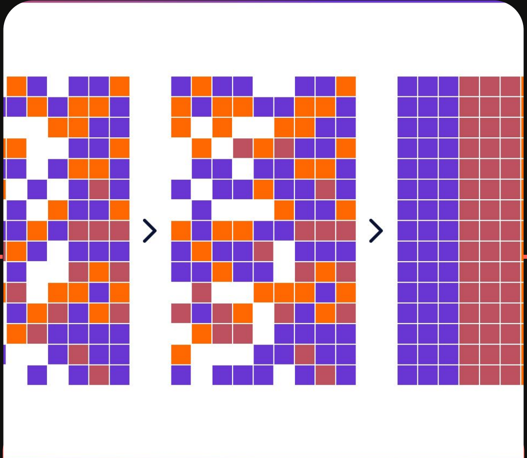
Pattern Recognition



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Example:
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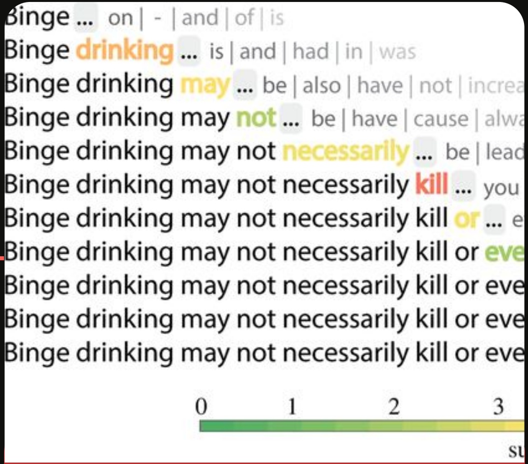
Generation Phase



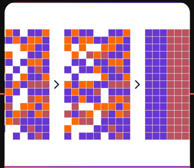
Predicts the best response using learned patterns

Example:
"What is metformin?" → Generates answer from all learned metformin information and data

Reading Phase

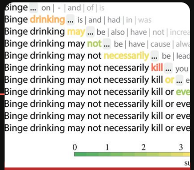
AI reads millions of books, websites, code repositories

Pattern Recognition



Builds connections between concepts and understands context

Generation Phase



Predicts the best response using learned patterns

The model is simply trained to predict next word

Learn to use AI tools for app building

Meet you AI prototyping tools

Replit

V0

Bolt

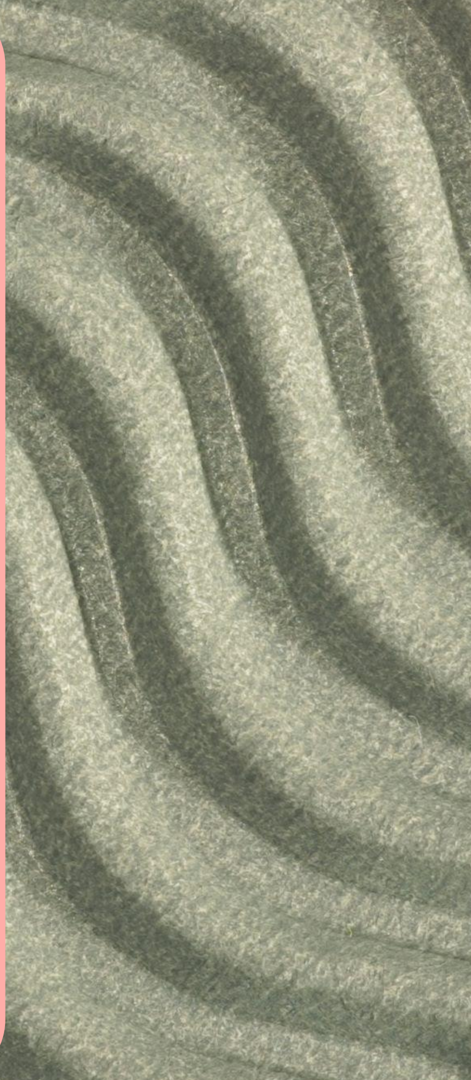
Lovable

Click and pick any of them today. They all work!

Free tier available.

Topic 3

Watch a live app being built from scratch



Quick Question?

What app do you wish to build the most?

It's All About the PROMPT

And understanding of course!

"Make me a pharmacy app" 👎 → Too vague, AI confused

"Create HTML code for a medicine information card" 👎 → More specific, but missing details

"Create a single page web app that displays medicine information in a card format. Include drug name, class, uses, dosage, and side effects. Use professional blue medical theme." 👍 → Clear, detailed, actionable

Good prompt follows the 5 Part Formula:

WHAT + FUNCTIONALITY + DATA + DESIGN + TECHNICAL

1. What

Define
Project Type

Why it matters
Sets the foundation for everything else

Example

'Create a web-based medicine information system'

2. Functionality

Define
List specific features and interactions

Why it matters
AI knows what to build

Example

"Users can search medicines by name or drug class"

"Display detailed drug information in organized cards"

"Allow filtering by therapeutic category"

What	Functionality	Data	Design	Technical

3. Data

Define
Specify what information to include

Why it matters
AI generates realistic, relevant content

or

Can also input ownself curated data

Example

"Database of 15 common Malaysian medicines"

"Each entry includes: generic name, brand names, dosage, indications, contraindications, side effects"

4. Design

Define
Describe visual style and UX

Why it matters
AI creates appropriate styling

Example

"Professional healthcare aesthetic with blue (#2563EB) and white theme"

"Card-based layout with clear typography"

"Mobile-responsive for phones and tablets"

5. Technical

Define
Technical requirement

Why it matters
AI chooses right technologies

Example

"Single-page application (SPA)"

"Static site (no backend required)"

"Works offline after initial load"

"Compatible with all modern browsers"

What	Functionality	Data	Design	Technical

Good and detail prompt

Full Prompt

Create a Pharmacy Drug Interaction Checker web application.

FUNCTIONALITY:
- Users select 2-3 medicines from dropdown menus
- Click "Check Interactions" button
- Display interaction severity (None/Minor/Moderate/Severe)
- Show detailed explanation of the interaction
- Provide recommendations for patients

DATA:
- Include 20 commonly prescribed medicines in Malaysia
- Pre-define known interactions between drug pairs
- Include severity levels and clinical significance

DESIGN:
- Clean medical interface with teal (#14B8A6) accents
- Traffic light colors for severity (green/yellow/orange/red)
- Large, readable fonts
- Warning icons for serious interactions
- Print-friendly layout

TECHNICAL:
- Pure HTML/CSS/JavaScript (no frameworks for simplicity)
- Embedded JSON data (no external API)
- Responsive design (mobile + desktop)
- Fast loading (<2 seconds)

What	Functionality	Data	Design	Technical

Create a [TYPE OF APP] for [AUDIENCE].

FUNCTIONALITY:
- [Feature 1]
- [Feature 2]
- [Feature 3]

DATA:
- [What information to include]
- [How much data]

DESIGN:
- [Visual style]
- [Color scheme with codes]
- [Layout preferences]

TECHNICAL:
- [Platform requirements]
- [Performance needs]
- [Browser compatibility]

Topic 4

Build your own personal app

Replit

V0

Bolt

Lovable

Wish don't come easy, let's get into action

Reality Check

LLMs are powerful, but they're not magic.

- ✅ Study tools & flashcard apps
- ✅ Drug information displays
- ✅ Simple calculators (dosage, BMI)
- ✅ Data collection forms
- ✅ Personal portfolio websites
- ✅ Medicine lookup tools

- ❌ Complex medical diagnosis systems
- ❌ Apps handling real patient data (privacy/security)
- ❌ Systems connecting to hospitals
- ❌ AI-powered drug discovery
- ❌ FDA-approved clinical tools
- ❌ Real-time patient monitoring

What Have We Accomplished Today

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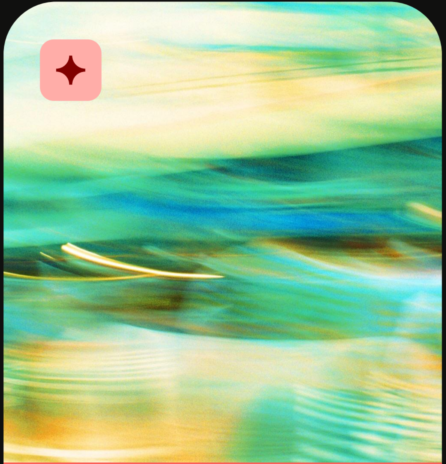

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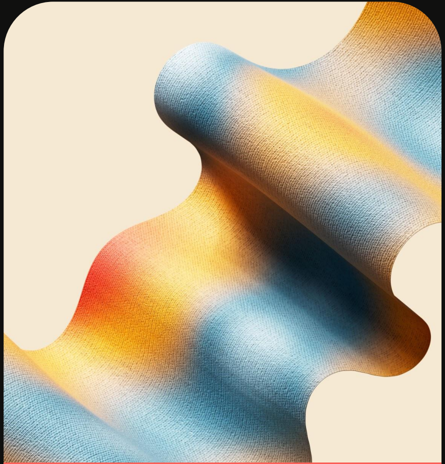

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Tailored | Curated

Bonus

GitHub.com

Education

Students Teachers Schools Events Sign in

GitHub Student Developer Pack

Learn to ship software like a pro. There's no substitute for hands-on experience. But for most students, real world tools can be cost-prohibitive. That's why we created the GitHub Student Developer Pack with some of our partners and friends.

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Discover the best ways to use pack offers with Experiences. Experiences are curated bundles of pack partner products, GitHub tools, and other resources that are designed for you learn new skills and make the most out of the Student Developer Pack and your journey in GitHub Education.

Thank you!